

Notes

All references are to the tenth edition.

1 Best Critical Regions

Neyman-Pearson helps us find a critical region at a given significance level α that has a lower chance of type II error than any other test at the same level—a “best critical region of size α ”. It states that if we have simple hypotheses $H_0 : \mu = \mu_0$ and $H_1 : \mu = \mu_1$, then any region of the form

$$C = \{(x_1, \dots, x_n) : L(\mu_0)/L(\mu_1) \leq k\},$$

where $k > 0$, is a best critical region of size α , where α is the size of C : $\alpha = P((X_1, \dots, X_n) \in C : \mu = \mu_0)$. Here $L(\mu)$ is the likelihood function:

$$L(\mu) = f(x_1, \dots, x_n; \mu) = f(x_1; \mu) \cdots f(x_n; \mu),$$

the joint pdf/pmf of the sample $(X_1, \dots, X_n)$.

Example (see also example 8.7-4): $X_1, \dots, X_n$ are i.i.d. $N(\mu, 9)$, and $H_0 : \mu = 10$, $H_1 : \mu = 20$. In this case

$$\begin{aligned} L(10)/L(20) &= \frac{(50\pi)^{-n/2} \exp(-\frac{1}{50} \sum_{i=1}^n (x_i - 10)^2)}{(50\pi)^{-n/2} \exp(-\frac{1}{50} \sum_{i=1}^n (x_i - 20)^2)} \\ &= \exp\left(-\frac{1}{50} \left(2(20 - 10) \sum_{i=1}^n x_i + n(10^2 - 20^2)\right)\right) \\ &= \exp\left(-\left(\frac{20 - 10}{25}\right) \sum_{i=1}^n x_i - (n/50)(10^2 - 20^2)\right). \end{aligned}$$

Now we want this expression to be small, specifically $\leq k$ for some $k > 0$. Hence, we want $\sum_i x_i$ to be big (since it is multiplied by the negative number $-(20 - 10)/25$, and then we exponentiate). So we want $\bar{x} = (1/n) \sum_i x_i$ to be at least as big as some fixed threshold c . Specifically, by taking logarithms, we see that $L(10)/L(20) \leq k$ is equivalent to

$$\bar{x} \geq \frac{(-50) \ln(k)}{2n(20 - 10)} + \frac{10 + 20}{2} = c.$$

Since $\ln(k)$ takes all possible values as k ranges over positive numbers, we can get any c on the right by choosing the right $k > 0$. So, we can choose c so that our test has the right significance level, and then by this computation we know that $\{\bar{x} \geq c\}$ is a best critical region for that level.

Moreover, notice that the same computation works if 20 is replaced by any $\mu_1 > 10$, because then $L(50)/L(\mu_1)$ becomes

$$\exp \left(- \left(\frac{\mu_1 - 10}{25} \right) \sum_{i=1}^n x_i - (n/50)(10^2 - \mu_1^2) \right).$$

Again, since the factor $-(\mu_1 - 10)/25$ is negative, we want $\sum_i x_i$ to be greater to make the whole exponential smaller. Taking logarithms we see that $L(50)/L(\mu_1) \leq k$ is the same as

$$\bar{x} \geq \frac{(-50) \ln(k)}{2n(\mu_1 - 10)} + \frac{10 + \mu_1}{2} = c.$$

Again, if we choose a fixed c as our threshold, we can find, for any $\mu_1 > 10$, a unique $k > 0$ solving this equation. Since $\mu_1 > 10$ was arbitrary, this means $\{\bar{x} \geq c\}$ is actually a “uniformly most powerful critical region” of its size, and the associated test is a “uniformly most powerful test”. So, choosing c so that we have the desired significance level, this test is actually best for the entire alternative hypothesis $H_1 : \mu_1 > 10$, not just $\mu_1 = 20$.

If on the other hand we wanted to test against $H_1 : \mu_1 < 10$, then the factor $-(\mu_1 - 10)/25$ would be positive, and so we would want $\bar{x}$ to be small. Hence, $\{\bar{x} \leq c\}$ is a uniformly best critical region of its size for testing against $H_1 : \mu_1 < 10$.

Although in this and many other examples, the critical region is defined in terms of the sample mean $\bar{X}$, this isn’t always the case, as in the following problem (solution at end):

Problem (exercise 8.7-8: Let $X_1, \dots, X_n$ be a sample from the pdf $f(x) = \theta(1 - x)^{\theta-1}$, $0 < x < 1$, $\theta > 0$ (a type of beta distribution). Find the form of the uniformly most powerful test of $H_0 : \theta = 1$ against $H_1 : \theta > 1$.

2 Sufficient Statistics

Usually, in estimating a parameter of a probability distribution, we don’t need the exact values $x_1, \dots, x_n$ of the sample. For example, in the section above, we only needed to know $\bar{x}$ in order to test hypotheses about μ . So, $\bar{x}$ is a “sufficient statistic” for estimating μ from a normal population. More generally, if the joint distribution of $X_1, \dots, X_n$ has the form

$$f(x_1, \dots, x_n; \theta) = \phi[u(x_1, \dots, x_n); \theta] \cdot h(x_1, \dots, x_n)$$

for some functions ϕ, u, h , then $u(X_1, \dots, X_n)$ is called a sufficient statistic for estimating θ . This is because, if the pdf/pmf factors like this, then the only interaction between θ and $(x_1, \dots, x_n)$ in the pdf is through the value $u(x_1, \dots, x_n)$. So, as far as estimating θ from $x_1, \dots, x_n$ is concerned, we may as well only focus on $u(x_1, \dots, x_n)$, and forget about the individual values $x_1, \dots, x_n$.

Example (exercise 6.7-7(a)): Let $X_1, \dots, X_n$ be random sample from the geometric distribution with parameter $0 < p < 1$: $f(x; p) = p(1-p)^{x-1}$, $x = 1, 2, \dots$. We will show that $Y = \sum_{i=1}^n X_i$ is sufficient for p . The joint pmf is

$$f(x_1, \dots, x_n; p) = p^n (1-p)^{-n + \sum_{i=1}^n x_i} = p^n (1-p)^{y-n} \cdot 1.$$

If $u(x_1, \dots, x_n) = x_1 + \dots + x_n = y$, $\phi(y; p) = p^n (1-p)^{y-n}$, and $h(x_1, \dots, x_n) = 1$, then we see from the definition that Y is sufficient for p .

Problem: let $X_1, \dots, X_n$ be a random sample from a gamma distribution with known $\alpha > 0$ and unknown $\theta > 0$. The pdf is $f(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}$, $x > 0$. (a) Show $Y = \sum_{i=1}^n X_i$ is sufficient for θ . (b) Show the maximum likelihood estimator of θ is a function of Y and is an unbiased estimator of θ .

3 Solutions

First Problem

We have

$$L(\theta) = f(x_1, \dots, x_n; \theta) = \theta^n (1-x_1)^{\theta-1} \dots (1-x_n)^{\theta-1} = \theta^n \left(\prod_{i=1}^n (1-x_i) \right)^{\theta-1}.$$

Therefore

$$L(1)/L(\theta) = \frac{1}{\theta^n \left(\prod_{i=1}^n (1-x_i) \right)^{\theta-1}} = \theta^{-n} \left(\prod_{i=1}^n (1-x_i) \right)^{1-\theta}.$$

So the desired region, by Neyman-Pearson, has the form $L(1)/L(\theta) \leq k$, or

$$\theta^{-n} \left(\prod_{i=1}^n (1-x_i) \right)^{1-\theta} \leq k,$$

or

$$\left(\prod_{i=1}^n (1-x_i) \right)^{1-\theta} \leq k\theta^n.$$

Now, because under H_1 , $\theta - 1 > 0$, the power on $\prod_{i=1}^n (1-x_i)$ is negative. Therefore, when we take the $1/(1-\theta)$ power of both sides of the inequality, it reverses:

$$\prod_{i=1}^n (1-x_i) \geq (k\theta^n)^{1/(\theta-1)} =: c.$$

Thus the uniformly most powerful test has critical region

$$C = \{(x_1, \dots, x_n) : \prod_{i=1}^n (1 - x_i) \geq c\}.$$

If H_1 was $\theta < 1$, then the inequality would become $\leq c$ instead, similar to the example with the normal distribution.

Second Problem

We have

$$f(x_1, \dots, x_n; \theta) = \frac{1}{\Gamma(\alpha)^n \theta^{n\alpha}} (x_1 \cdots x_n)^{\alpha-1} e^{-\frac{1}{\theta} \sum_{i=1}^n x_i} = (\theta^{-n\alpha} e^{-y/\theta}) \cdot \frac{(x_1 \cdots x_n)^{\alpha-1}}{\Gamma(\alpha)^n}.$$

If $u(x_1, \dots, x_n) = x_1 + \dots + x_n = y$, $\phi(y; \theta) = \theta^{-n\alpha} e^{-y/\theta}$, and $h(x_1, \dots, x_n) = \frac{(x_1 \cdots x_n)^{\alpha-1}}{\Gamma(\alpha)^n}$, then we see from the definition that Y is sufficient for θ .

Note: it is also possible to do this and the above example using exponential families; see section 6.7.